

Assignment 2

1 Requirements

You are expected to complete all required exercises and encouraged to complete the optional ones. For submission, please put all your answers in a PDF file. The name of the file should follow the format “**studentID_assignmentNumber**” (e.g., 2025111054_assignment2). **The submission deadline is 11:59 PM, October 16, 2025**. Please ensure that your answers (including NFAs and DFAs) are clear enough, otherwise it will affect the evaluation. It is recommended to use drawing software to draw NFAs and DFAs.

2 Required Exercises (100 points)

Exercise 1: Design NFAs and DFAs to recognize each of the following regular languages. Detailed steps can be omitted.

1. $L((a|b)^*b)$ [10 points]
1. $L(((\epsilon|a)^*b)^*)$ [10 points]
2. $L((a|b)^*a(a|b)(a|b))$ [10 points]
3. $L(a^*ba^*ba^*ba^*)$ [10 points]

Exercise 2: Convert the following regular expressions to NFAs using the **Thompson's Construction Algorithm** (Algorithm 3.23 in the dragon book). **Please put down the detailed steps.**

1. $((\epsilon|a)^*b)^*$ [10 points]
1. $(a|b)^*a(a|b)(a|b)$ [10 points]
1. $a^*ba^*ba^*ba^*$ [10 points]

Exercise 3: Convert the NFAs in Exercise2 to DFAs using the **Subset Construction Algorithm** (Algorithm 3.20 in the dragon book). **Please put down the detailed steps.** [30points in total; 10 points for each correct conversion]

3 Optional Exercises (20 bonus points)

Exercise 1: Minimize the number of states of the DFAs you have built for regular expressions 2 and 3 in Exercise 2 using the State-Minimization Algorithm (Algorithm 3.39 in the dragon book). **Please put down the detailed steps.** [10 points for each correct minimization process]